

Khmelnyskyi

Adaptive Traffic Management Pilot Project

Bringing Research into Urban Governance Practice

267 891

residents

20,6 км²

city area

Pilot Theme: Adaptive Traffic Management



Problem: Static traffic management systems do not respond to real road conditions, do not account for city terrain, and do not optimize flows in real time - leading to congestion and elevated CO₂ emissions.



Smart Sensors and Cameras

Real-time traffic data collection from intersections



Artificial Intelligence

Pattern recognition and adaptive management algorithms



CO₂ Reduction

Flow optimization - fewer stops and unnecessary emissions

Current State: Outdated Infrastructure



Khmelnytskyi's traffic management system is technically obsolete. Integrating AI into it is impossible without replacing the hardware.

NOW — What We Have

- Fixed signal cycles - no adaptation
- No real-time sensors and and problematic data collection
- No integration between control nodes
- Impossible to connect AI algorithms
- City terrain is not taken into account



NEEDED - for AI Management

- ✓ Controllers with open API
- ✓ Sensors and cameras at intersections (cameras exist, but their software must be integrated with traffic management systems)
- ✓ Communication network between nodes
- ✓ Real-time data processing platform
- ✓ Topographic module (terrain)

Scientific Foundation: What Has Been Done



Paper 1 - Future Transportation, 2025

Adaptive ML approach to traffic pattern recognition (HDBSCAN + k-means)

0.79–0.82

V-measure

95%

Identification Accuracy

0.94

Temporal Coherence



Paper 2 - Future Transportation, 2026

Topography-aware deep learning for traffic management (DRL + CSRD)

–8.6%

CO₂ vs Max-Pressure

–4.7%

CO₂ vs standard DQN

Both papers validated on a digital twin of Khmelnytskyi's transport network (SUMO simulation, 15 intersections, 45.7 km)

What We Do at Each Stage

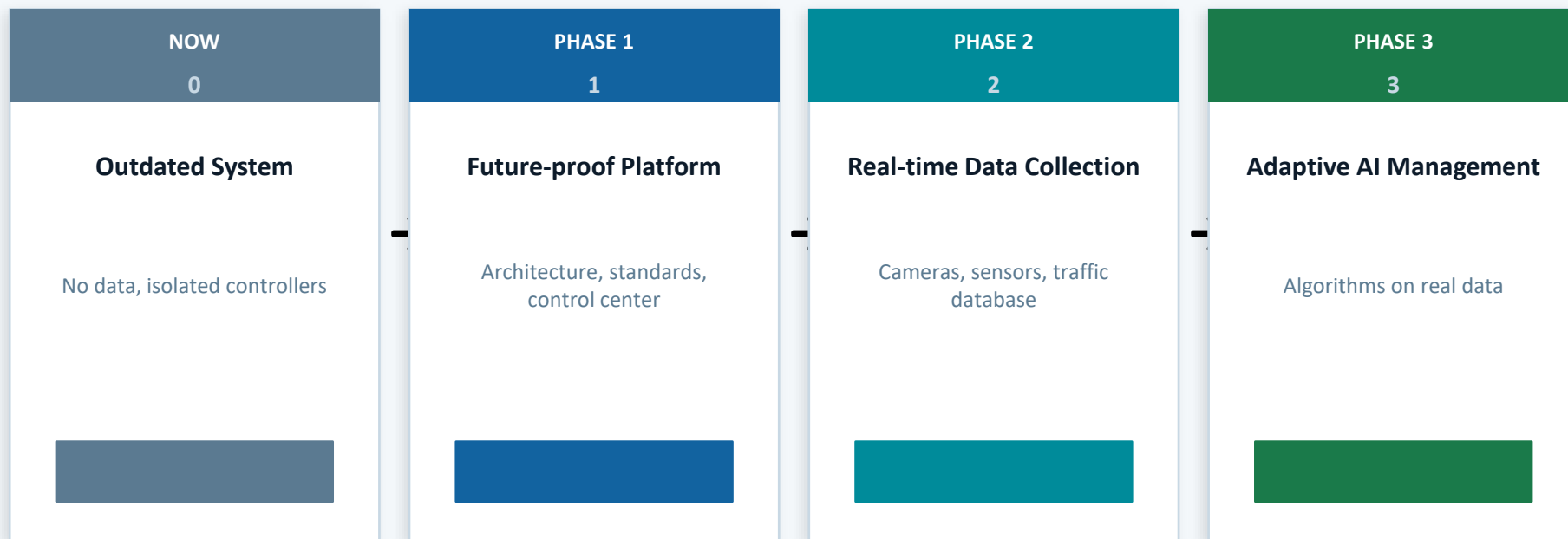
Phase	What We Do	Outcome
0 · Now	Document current state, define the problem, formulate technical specifications	Technical Specifications
1 · Platform	Unified digital platform · connectivity standards Control center · secured network	Future-proof Foundation
2 · Data	Cameras and counters at key intersections Database of traffic intensity, flow composition, peak hours	Real Database
3 · Pilot	Algorithms at 3–5 intersections Measuring actual CO ₂ reduction and trip duration	First Real Results
4 · Scale	Expansion to all intersections · Management Center Model for other cities in Ukraine and the EU	Full AI System

💡 Simulations → real data: when Phase 2 data becomes available, the algorithms from the research papers will become significantly more accurate

Roadmap: Foundation First - Then the Building



Direct AI deployment is impossible: no modern infrastructure or real data. Logic - foundation first, then algorithms.



Research - a ready-made tool for Phase 3, waiting for the right foundation

Strategy: Gradual Modernization Informed by Research



Principle: with any replacement or repair - install AI-compatible equipment. Not all at once, but right from the start.

01 Pilot Zone (after Phase 2)

- ✓ Select 3–5 key intersections
- ✓ Install modern controllers + cameras
- ✓ Connect algorithms
- ✓ Demonstrate real CO₂ reduction

02 Expansion (after results)

- ✓ Replicate experience to new nodes
- ✓ Replace old equipment as it wears out
- ✓ Integrate with municipal systems
- ✓ Attract new funding based on results

03 Unified System (long-term)

- ✓ City Traffic Management Center
- ✓ All nodes in a unified digital network
- ✓ AI manages the city in real time
- ✓ Model for other cities in Ukraine and the EU

How to Embed Research into City Plans

What to Include in the Plan

City Transport Plan

AI-compatibility requirement for new equipment

Recovery Plan

Transport modernization with smart technologies

Procurement Standards

New controllers - only with open API

KhNU – Council-Partnership

University as a permanent technical partner

Why This Is Realistic



Research is published - technical foundation is ready



No need to replace everything at once - as equipment wears out



Future-proof approach - meets EU requirements



Results of each step - an argument for grants



KhNU remains a partner - knowledge is not lost

Key Uncertainties for Scenario Planning



Wartime Context

Infrastructure destruction, blackouts, unpredictability of planning



Community Acceptance

Will residents support new restrictions and traffic changes?



Outdated Infrastructure

Full equipment replacement is needed - when and where will the funds come from?



Data and Reliability

Will there be a technical base for the system to operate under crisis conditions?



Financial Sustainability

Municipal budget, EU grants, donors - what will be available?



Scalability

Can the model be transferred to other cities in Ukraine and the EU?

Our Request for Today's Workshop



Identify which uncertainties are most critical for the pilot



Outline key stakeholders and vulnerable groups



Assess the city's readiness - city's -technical, institutional, financial



Together formulate under what conditions implementation will succeed

The technology exists. The research exists. The question is - in what kind of future will we implement them.